

BSc(Surveying) | RECO40II Dissertation

Mapping Hong Kong's Low-Altitude Economy Policies
under Unmanned Aerial Vehicle Adoption in Site Inspection

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Major Findings

1. Financial barriers (High investment cost and high operational cost) and technical barriers (High software requirements and high hardware requirements) are the root barriers to adopting UAV site inspection.
2. The current UAV policy fails to provide specialised training, technical performance standards, construction-site-specific operational flexibility, and financial support.
3. Major targeted policy recommendations include the establishment of construction site-specific operation guidelines and incentives for drone service outsourcing.



Abstract

The adoption of unmanned aerial vehicles (UAVs) for site inspection in Hong Kong's architecture, engineering, and construction (AEC) sector has the potential to significantly enhance efficiency, safety, data quality, and cost-effectiveness. However, despite technological advancements and strong government promotion for the Low-Altitude Economy (LAE), the actual adoption rate remains limited and fragmented.

This study systematically identified and analysed 14 key barriers to UAV adoption using Interpretive Structural Modelling (ISM). The results revealed that financial and technical barriers are the primary root barriers, exerting the strongest influence on the system. Educational, safety, environmental, and regulatory barriers were found to be dependent on these root causes.

By mapping the identified barriers against Hong Kong's Small Unmanned Aircraft Order (Cap. 448G) and the LAE Regulatory Sandbox, the study proposed a series of targeted policy recommendations. These include the establishment of a certification framework for approved software and hardware, a subsidy scheme for UAV investment and outsourcing, a dedicated Technical Performance Track in Sandbox X, specialised AEC training programmes for pilots, construction site-specific operation guidelines, and incentives for drone service outsourcing.

The findings highlight the need to shift from a technology-centric to a barrier-driven approach in LAE policy development. This research contributes both theoretically by demonstrating the application of ISM in emerging technology policy analysis and practically by offering recommendations to policymakers and industry stakeholders to foster safe and effective UAV adoption for construction site inspection in Hong Kong.

